

Additional text 3: is there one axis shared by all the responses?

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Question

The main report found recurring classes: membrane proteins rise with salt in the HSPB1 pull-down, prefer DNAJB11 over DNAJA1, and fall with heat; disordered, charged proteins tend the other way. Is there one axis of protein properties behind all the responses, and how much of each response is left once it is removed?

Data and method

6,331 proteins measured in both experiments and kept for all eight reproducible responses: salt (75 and 150 mM, HSPB1), heat (37 and 43 vs 35 °C), staurosporine (35, 37, 43 °C), and the DNAJB11-versus-DNAJA1 preference; the GA_33 responses averaged over the two co-chaperones.

Correlations between responses need care. The salt and GA_33 experiments are independent, so their correlation is corrected for measurement noise with each response's reliability in the usual way. Within GA_33 the responses reuse the same control samples (heat at 43 °C is control 43 minus control 35; staurosporine at a temperature is treated minus control at that temperature), so their measurement noise is shared and would bias their correlation, in a direction set by that arithmetic. Within GA_33, therefore, each response was computed separately from biological replicates 1-3 and 4-6, and the correlation of one response from one half with the other from the other half was used, corrected for the reliability of a three-replicate estimate. The two salt doses also share their 0 mM samples, but measurement noise is about 1 % of the salt variance, so this is negligible there. Intervals resample sequence clusters.

Results

1. The responses are largely independent of one another. Correlations, free of shared noise and corrected for measurement noise:

	salt 75	salt 150	heat 37	heat 43	stau 35	stau 37	stau 43	DNAJB11 pref.
salt 75	1							
salt 150	+0.92	1						
heat 37	-0.17	-0.13	1					
heat 43	-0.27	-0.24	+0.78	1				
stau 35	+0.01	+0.03	+0.10	-0.04	1			
stau 37	-0.04	-0.04	-0.11	-0.01	+0.92	1		
stau 43	+0.02	+0.02	-0.02	-0.07	+0.79	+0.76	1	
DNAJB11 pref.	-0.05	-0.04	+0.31	+0.29	-0.23	-0.11	-0.08	1

The strong correlations are within a perturbation: the two salt doses, the two heat temperatures, and the three staurosporine temperatures. Between perturbations the only appreciable links are salt with heat (-0.24 [-0.27,

-0.21] for 150 mM and 43 °C) and heat with the DNAJB11 preference (+0.28 [+0.26, +0.30] from disjoint halves). Staurosporine is unrelated to salt and to heat. The principal components of the corrected matrix are, in order, the staurosporine response (34 %), salt against heat (29 %), and a mixture of salt, heat and preference (19 %).

2. The other experiment explains little of each response. Cross-validated over sequence clusters, the responses of one experiment predicting each response of the other:

Response	Explained by the other experiment	ESMC on the response	ESMC on what is left
Salt 75 mM	7.4 %	17.0 %	13.4 %
Salt 150 mM	6.4 %	18.5 %	14.7 %
Heat 37 °C	2.8 %	11.2 %	9.5 %
Heat 43 °C	7.1 %	21.8 %	18.3 %
Staurosporine 35/37/43 °C	0 %	5.0-7.1 %	4.9-7.0 %
DNAJB11 preference	0 %	44.3 %	44.3 %

Most of what sequence predicts for each response is specific to that response.

3. The salt-heat link is explained by each protein’s level in the two pull-downs, not by a shared protein property. The link does follow protein class: membrane proteins sit at one end (rise with salt, fall with heat; class mean +0.83 on the standardised salt-minus-heat score, $p = 2e-103$), proteins found only in the cytoplasm at the other (-0.38), and it persists without membrane proteins (-0.21). But each response depends on how much of a protein its own pull-down contains, and the two pull-downs’ contents are correlated (+0.59 across proteins). What drives the link is a protein’s level in the HSPB1 pull-down relative to the DNAJ pull-downs, which correlates -0.53 with the salt response and +0.53 with the heat response:

Salt 150 mM vs heat 43 °C	Correlation
Raw	-0.241
Each adjusted for its own pull-down’s level	-0.107
Both adjusted for both pull-downs’ levels	-0.013

Proteins relatively enriched in the HSPB1 pull-down lose HSPB1 association with salt and gain DNAJ association with heat, and proteins relatively enriched in the DNAJ pull-downs do the reverse. Each perturbation erodes what is specifically enriched in its own pull-down, and that, not a shared sequence property, is what links the two experiments.

4. The class effects within each experiment are not abundance effects. Removing each response’s dependence on its own pull-down level leaves the sparse-autoencoder readings of the main report in place:

Response	Feature or class	Raw	Abundance removed
Salt 150 mM	Transmembrane helix exit signature (SAE 9620)	+0.190	+0.186
	Hydrophobic membrane helices (SAE 11444)	+0.181	+0.174
	Transmembrane proteins vs others (log2)	+0.46	+0.39
Heat 43 °C	Long low-complexity IDRs (SAE 654)	-0.189	-0.291
	Mixed-charge low-complexity IDRs (SAE 10715)	-0.164	-0.237
	Basic/polar N-terminal segments, enzymes (SAE 5338)	+0.195	+0.224
	Alpha-helical transmembrane segments (SAE 5659)	-0.177	-0.128
	Transmembrane proteins vs others (log2)	-0.52	-0.34

The disorder effect on the heat response is stronger once abundance is removed; part of the membrane effect on heat is abundance, but most of it remains.

5. A sequence-only membrane-versus-disorder axis tracks one response. Scoring each protein by the mean activation of 107 transmembrane features minus that of 90 disorder and low-complexity features (chosen

by their Biohub category and label, not by the data), the score correlates +0.43 with the DNAJB11 preference, +0.10-0.12 with salt, and about 0 with heat and staurosporine. With heat both membrane and disordered proteins fall, so a membrane-minus-disorder contrast cancels.

Interpretation

There is no single protein axis behind the four perturbations. Each response has its own sequence determinants (membrane helices for salt, folded-versus-disordered for heat, kinase domains for staurosporine, secretory-versus-nuclear for co-chaperone preference), and they overlap little. The one link between experiments, salt against heat, is explained by how each protein is distributed between the two pull-downs: a perturbation removes what its own bait specifically enriched. That is worth knowing for any comparison across these pull-downs, because it will reappear wherever two pull-downs with different baits are compared.

Critical evaluation

- **Shared samples.** The GA_33 responses reuse the same control samples. Using disjoint replicate halves removes the shared noise but leaves only three replicates per estimate, and the correction for that divides by a half-replicate reliability (0.4-0.98), which inflates the noisier staurosporine correlations. The intervals quoted for the GA_33 pairs are before that correction.
- **Abundance as an explanation.** “Level in the pull-down” mixes cellular abundance with binding to the bait. The input lysate would separate the two; until then, “each perturbation erodes what its bait enriched” is the most direct reading, not a demonstrated mechanism.
- **The abundance adjustment is quadratic.** A more flexible adjustment could remove a little more, but the salt-heat correlation is already at -0.01.
- **Coverage.** Only proteins measured in both experiments and kept for all eight responses enter; this excludes the least abundant and least reproducibly measured proteins.

Methods files

`code/ad03_shared_axis.py` (correlations, components, the sequence-only axis), `code/ad03b_shared_axis.py` (shared-noise-free correlations, cross-experiment prediction, the salt-heat axis and its abundance analysis); results in `data/ga_data/ad03_results.json` and `ad03b_results.json`.